Optimizing Maintenance Routes for Highway Infrastructure Using Leader-Follower Autonomous Vehicles

Qing Tang, Chenxi Chen, Xianbiao Hu, Yuxin Ding, Tianjia Yang

Abstract

The emerging technology of the Autonomous Truck Mounted Attenuator (ATMA), a leader-follower style vehicle system, leverages connected and automated vehicle capabilities to enhance safety during transportation infrastructure maintenance in work zones. However, ATMA vehicles, due to their significantly lower speed compared to regular vehicles, create moving bottlenecks that reduce roadway capacity and increase queue lengths, leading to additional delays. The different routes taken by ATMA result in varied patterns of time-varying capacity drops, which can affect the user equilibrium traffic assignment and lead to different system costs. This manuscript aims to optimize the routing for ATMA in a network to minimize the system cost associated with their slow-moving operation.

To achieve this, a queuing-based traffic assignment approach is proposed to quantify the system cost incurred by ATMA on a specific route. A queuing-based time-dependent (QBTD) travel time function, which considers capacity drop, is introduced and applied to the static user equilibrium traffic assignment problem (TAP), adding dynamic characteristics. The queuing-based TAP is then formulated and solved using a modified path-based algorithm. The methodology is validated using both small and large networks and is compared with two benchmark models to assess the benefits of capacity drop modeling and the QBTD travel time function. Furthermore, the approach is applied to evaluate the impact of different routes on the network and to identify an optimal route for ATMA during maintenance work. Finally, sensitivity analysis is conducted to explore how the impact changes with variations in traffic demand and capacity reduction.

Keywords: Autonomous Truck Mounted Attenuator; Capacity Drop; Queuing-based Time-dependent Travel Time Function; User Equilibrium; Traffic Assignment

Abbreviations

ATMA Autonomous Truck Mounted Attenuator
BPR Bureau of Public Roads
CAV Connected and automated vehicle
DTA Dynamic traffic assignment
FD Fundamental diagram
FFTT Free-flow travel time
QBTD Queuing-based time-dependent
STA Static traffic assignment
TAP Traffic assignment problem
TSTT Total system travel time
UE User equilibrium
UETAP User equilibrium traffic assignment problem

1. Introduction

The Autonomous Truck Mounted Attenuator (ATMA) system combines the capabilities of connected and automated vehicle (CAV) with Autonomous Maintenance Technology to maintain transportation infrastructure in work zones. This leader-follower system, featuring a manned leader and an unmanned follower equipped with a truck-mounted attenuator, eliminates the need for workers in the follower, significantly reducing fatalities [1]. The leader vehicle performs maintenance work, while the follower mirrors its actions and serves as a protective buffer, both equipped with actuators, software, electronics, and vehicle-to-vehicle communication for autonomous driving. The Federal Highway Administration and several State DOTs, including Colorado and Missouri, have strongly supported and pioneered ATMA deployment [2]. Research has assessed ATMA's performance and operational design domain to guide DOTs on optimal deployment locations and timing based on traffic performance [3, 4].

However, an optimal network deployment plan has not yet been established. This research aims to identify the best routing strategy for ATMA to minimize efficiency loss across the entire transportation system. We assume that multiple links require maintenance, with ATMA starting and returning to the same location. ATMA typically operates at slow speeds (5-15 mph) due to the nature of mobile, slow-moving operations. This significant speed difference compared to regular traffic creates a moving bottleneck, resulting in reduced roadway capacity [3, 4]. During maintenance operations, this capacity drop causes additional delays and affects travel time on the roadway. When the maintenance schedule is announced in advance, general traffic may divert to alternative routes, establishing a new user equilibrium (UE). The presence of ATMA on various routes impacts system costs, such as total system travel time (TSTT). This study's goal is to quantify these system costs and determine the optimal routing plan.

The primary challenge is performing UE traffic assignment while accounting for time-varying capacity reduction to quantify system costs. The user equilibrium traffic assignment problem (UETAP), introduced by Beckmann et al. [5], is generally categorized into static traffic assignment (STA) and dynamic traffic assignment (DTA), differentiated by their traffic flow models used. STA

models rely on link performance functions, such as the Bureau of Public Roads (BPR) function, making it relatively easy to find UE even for large-scale networks. The static UETAP can be mathematically formulated, ensuring the existence and uniqueness of equilibrium solutions under mild conditions. Several algorithms have been developed to solve the problem, including the Method of Successive Averages [6], Frank-Wolfe algorithm [7], Gradient Projection Method [8], and Bush-based or origin-based algorithm [9]. STA models are favored for traffic planning due to their computational efficiency, low input requirements, robustness, tractability, and accountability [10]. However, they lack accuracy in congested conditions, failing to account for flow metering, spillback effects, and variability in travel times across vehicles on a link [11, 12]. As such, traditional STA models are not applicable for the traffic assignment when considering the ATMA moving bottleneck.

Over the past decades, significant efforts have been made in developing DTA models to predict traffic conditions. These models capture realistic traffic flow and driver responses by tracking time-varying link flows and travel times, enabling more accurate route choice determination. DTA models are broadly categorized into analytical-based and simulation-based approaches [13], with simulation-based methods being more commonly used due to their flexibility in network loading, ability to simulate traffic flow propagation, and capacity to capture spatial and temporal interactions while determining travel costs. Time-dependent travel times are typically generated using macroscopic [14, 15], mesoscopic [16, 17], or microscopic [18, 19] models. Mesoscopic simulations, often used in the network loading module, provide a resolution of 5 to 10 seconds [20] and are employed in tools such as DYNASMART [16], DYNAMIT [17, 21], and DTALite [22]. However, DTA models generally lack clear mathematical properties and convergence guarantees, limiting their practical applicability [20, 23]. Moreover, simulation-based models, often integrating mesoscopic simulators into macroscopic models, may introduce inconsistencies between resolution levels, leading to unrealistic oversaturated conditions [24].

To address these limitations, this study proposes a novel queuing-based time-dependent (QBTD) traffic assignment approach, introducing dynamic characteristics into the traditional static UETAP to optimize ATMA routes. The key contributions are outlines as follows:

- A QBTD travel time function is proposed, incorporating capacity drop caused by ATMA's moving bottleneck. This approach bridges the gap between static and dynamic traffic assignment models, offering a balance between accuracy and computational efficiency by better capturing traffic flow dynamics and congestion propagation.
- The TAP is reformulated to account for dynamic traffic flow characteristics. A modified path-based algorithm is developed to solve the problem, achieving a dynamic UE.
- The study quantifies the influence of ATMA routes on total system travel time and develops a bi-level optimization framework to determine the optimal routing strategy. The higher-level objective aims to minimize TSTT by identifying the most effective ATMA routes, while the lower-level problem evaluates system costs through UE traffic assignment, accounting for time-varying capacity reductions.
- The research provides a framework for optimizing ATMA deployment in transportation infrastructure maintenance, promoting safer and more efficient work zone operations. The findings support DOTs in making informed decisions about route selection to minimize traffic disruptions.

The rest of this manuscript is organized as follows: Section 2 presents the preliminaries of the capacity drop induced by the moving bottleneck and the QBTD travel time function. The overall module with bi-level objectives is summarized in Section 3. In Section 4, numerical experiments are conducted to compare the performance with two common-used models and to investigate the quantitative impact of different routes on the traffic system. Sensitivity analysis is also performed to explore how the impact of ATMA changes. Section 5 concludes this paper.

2. Preliminaries

2.1. Problem Statement

Assuming there is a set of links in a generic network that require maintenance, a routing plan must be scheduled in advance. The ATMA should leave from the depot location, sequentially maintain the links, and eventually return to the same depot location. Typically, drivers select the shortest path to their destination. However, ATMA vehicles, which generally travel at speeds lower than 15 mph to perform maintenance, act as a moving bottleneck that reduce roadway capacity [3, 4]. This reduction leads to increased queue lengths and delays, degrading the level of service of the affected links. Moreover, the increased delays may cause other vehicles to detour to alternative routes, resulting in disrupting the original UE pattern. The objective is to identify the optimal route that maintains the links in a specific sequence, minimizing the impact of maintenance. In the following sections, we introduce the capacity drop induced by ATMA moving bottleneck and a QBTD travel time function.

2.2. ATMA Moving Bottleneck and Capacity Drop

Figure 1 describes a typical highway with two lanes in the same direction. The ATMA is represented by the two vehicles within the red box, consisting of a leader vehicle and a follower vehicle, with a gap distance of L_{gap} between

them. Other general vehicles are represented by smaller black vehicle icons. The speed of the ATMA, denoted as v_{ATMA}, is typically lower than the cruising speed v_{u}. Blocked by the ATMA moving bottleneck, the vehicle behind it will attempt to bypass it using the left lane, i.e., the red vehicle.

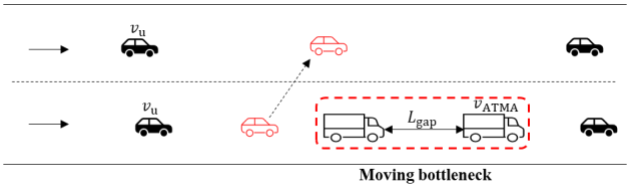


Figure 1. Schematic diagram of a four-lane highway (one direction) segment with ATMA vehicles.

Newell [25] suggests that, through appropriate coordinate transformations, a moving coordinate system can be analyzed as flow passing a stationary bottleneck. By positioning a "moving observer" on the main road traveling at the same speed as the ATMA, the two-lane roadway segments appear, from the observer's perspective, as a stationary section with only one lane.

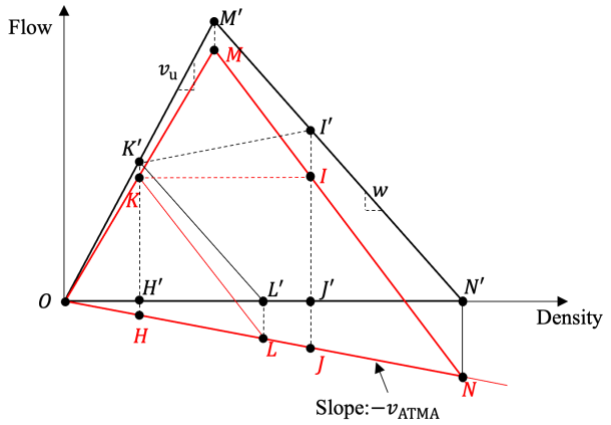


Figure 2. Flow-density relationship in the moving and stationary coordinates.

We refer to the perspective of the moving observer as the moving coordinate and that of the stationary observer as the stationary coordinate. From the stationary observer's perspective, the reduced capacity is the maximum discharge rate. Figure 2 presents q-k fundamental diagrams (FD) of a one-lane and a two-lane roadway. The stationary coordinate is marked in black ($OM'N'$) and the moving coordinate is marked in red (OMN). Specifically, triangles $OM'N'$ and $OK'L'$ represent the FD of two-lane and one-lane segments, respectively. Here, v_u is the cruising speed is, w is the backward wave speed, and k_j is the jam density. By deducting a flow $k \cdot v_{\mathrm{ATMA}}$ from triangle $OM'N'$, the resulting FD in the moving coordinate are OMN and OKL.

In the stationary coordinate system, $K'H'$ is the maximum discharge rate of the one-lane segment. After a reduction by $OH' \cdot v_{\mathrm{ATMA}}$, it becomes KH', the maximum discharge rate in the moving coordinate system. The downstream bottleneck traffic state is represented by point K, while the upstream state at point I, which has the same outflow rate as point K but has a higher density due to the queue behind the moving bottleneck. Converting back to stationary coordinates, points K and I become points K' and I' by adding a flow of $k \cdot v_{\mathrm{ATMA}}$. Point I' determines the reduced capacity μ' in the stationary coordinate with the ATMA moving bottleneck, shown as the length $I'J'$ derived by the Eq. (1). For detailed derivations, please refer to Tang and Hu [3, 4] for further information.

$$\mu' = \mu \cdot \frac{2v_{\mathrm{u}} \cdot v_{\mathrm{ATMA}} + v_{\mathrm{ATMA}} \cdot w + w \cdot v_{\mathrm{u}}}{2(v_{\mathrm{ATMA}} + w) \cdot v_{\mathrm{u}}} \tag{1}$$

2.3. Queuing-based Time-dependent Travel Time Function

While the ATMA maintains a road link, its travel time function is impacted by a capacity drop. This section introduces a QBTD travel time function that accounts for this effect. In uncongested conditions, the aggregated vehicle speed on each link remains steady and is approximated using free-flow speed [26]. In congested states, as vehicle speed decreases due to queuing, travel time includes both free-flow travel time (FFTT) and queue delay. Queue delay is time-varying, influenced by changing traffic demand or capacity. Based on Newell's fluid approximation for queue dynamics[27], we consider both uncongested and congested states.

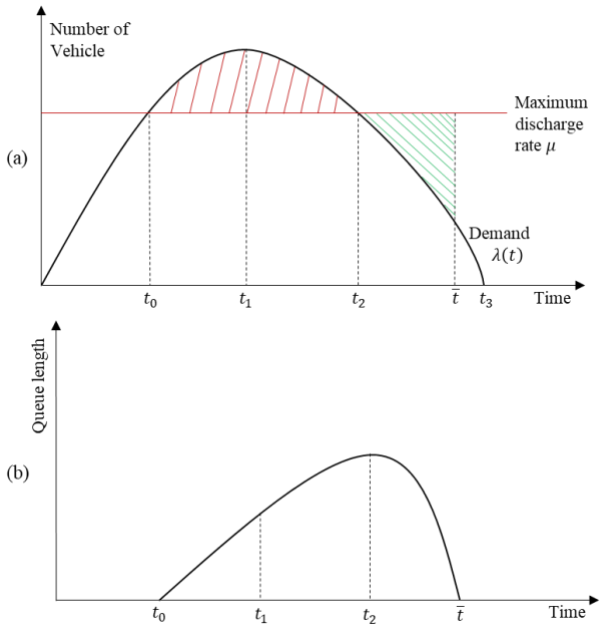


Figure 3 (a) illustrates the demand-supply relationship: from t_0 to t_2, demand $\lambda(t)$ (black curve) exceeds the maximum discharge rate μ (red line), creating a queue, shown in Figure 3 (b). As arrival rates fall below μ, the queue dissipates, fully clearing by time $\bar{t}$. The time-varying queue length at any given time t is then calculated as:

$$Q(t) = \int_{t_0}^{t} (\lambda(\tau) - \mu)\, d\tau \tag{2}$$

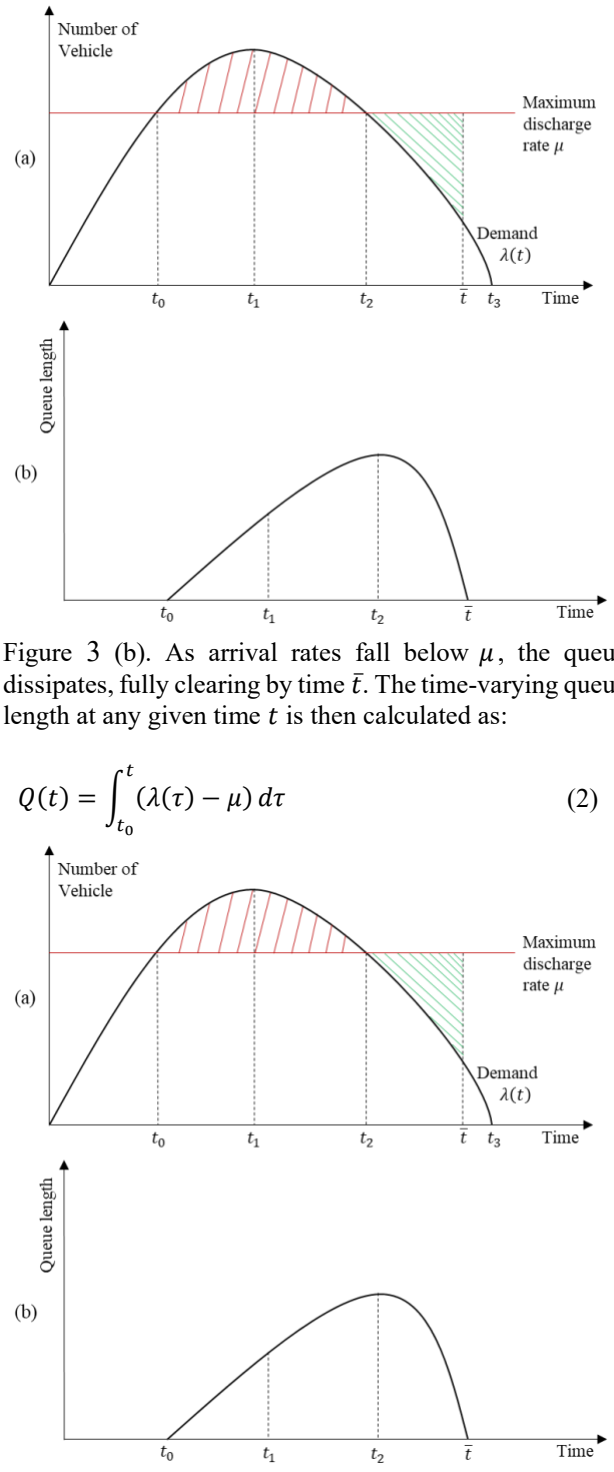


Figure 3. Queue formation and dissipation processes: (a) time-varying demand curve; (b) queuing profile.

Figure 4(a) illustrates the cumulative arrival curve $A(t)$ and departure curve $D(t)$. To describe the queue at time t, we shift the cumulative arrival curve by the FFTT to obtain a virtual arrival curve $V(t)$. Hence, the vertical distance between $V(t)$ and $D(t)$ represents the queue length $Q(t)$, and the horizontal distance corresponds to the time-dependent delay:

$$w(t) = \frac{Q(t)}{\mu} \tag{3}$$

As mentioned previously, the travel time consists of the FFTT plus the time-dependent delay. For a general purpose, the QBTD travel time for a specific link in a network can be derived as:

$$TT(t) = t^f + w(t + t^f) \tag{4}$$

where t represents the departure time or the arrival time at the upstream of a link, t^f denotes the FFTT for a link, $w(t + t^f)$ the delay with the departure time t.

Proposition 1. The QBTD travel time function in Eq. (4) satisfies the FIFO property.

Proof. The FIFO property can be proved if the inequality $t_1 + TT(t_1) \le t_2 + TT(t_2)$ holds for any link and any time $t_1 \le t_2$. Let $f(t) = t + TT(t)$, in which t is the departure time, and $TT(t)$ is the link travel time as the departure time is t.

$$\frac{df(t)}{dt} = \frac{\lambda(t + t^f)}{\mu} \tag{5}$$

If $t_0 \le t \le t_2$, it is easy to observe that $\frac{df(t)}{dt} \ge 1$; otherwise, it can be found that $0 \le \frac{df(t)}{dt} < 1$. Thus, function $f(t)$ is non-decreasing, indicating that the link travel time function $TT(t)$ satisfies the FIFO property.

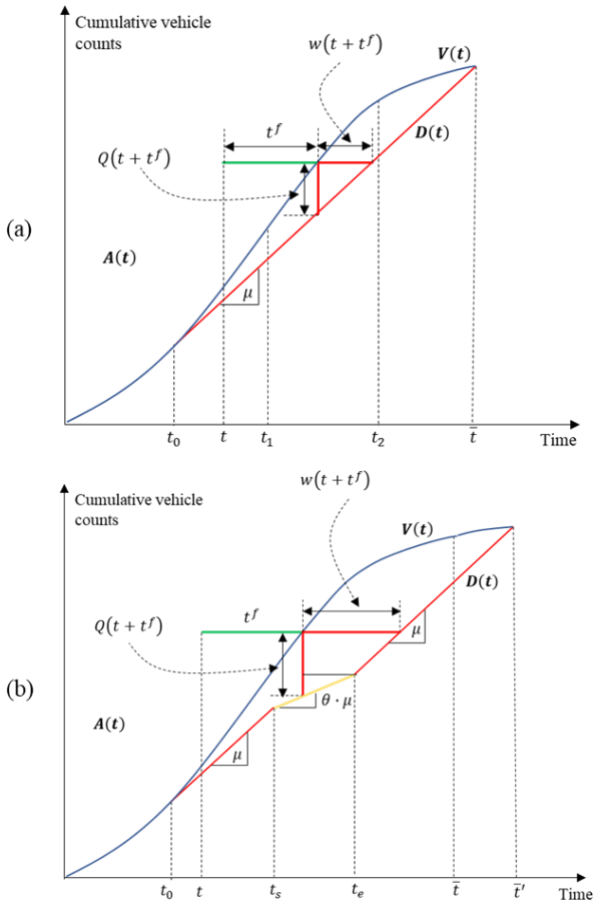


Figure 4. Cumulative curves and time-dependent travel time with (a) constant capacity and (b) discounted capacity.

Next, let's examine how the reduced capacity drop μ' affects the QBTD travel time function during ATMA vehicles maintaining the link. Figure 4(b) describes the cumulative arrival and departure curves that account for the reduced capacity as the ATMA vehicles traverse within a specific period $[t_s, t_e]$. Consequently, the queue still exists at time $\bar{t}$ and is discharged at time $\bar{t}'$. The time-dependent delay can be derived for different departure times at the upstream of a link. For example, if $t_s \le t + t^f \le t_e$, the delay time is derived as Eq. (6). The FIFO property is still satisfied with the corresponding QBTD travel time function with the capacity drop.

$$w(t+t^f) = (t_e - t - t^f) + \frac{Q(t+t^f) - (t_e - t - t^f)\cdot \mu'}{\mu} \quad (6)$$

3. ATMA Routing Optimization

In this section, we begin by defining the system cost due to the capacity drop during maintenance. A modeling framework with bi-level objectives is proposed to identify an optimal routing. This framework encompasses a higher-level objective which focuses on finding an optimal ATMA route with the lowest system cost and a lower-level objective which involves calculating the system cost by conducting UE traffic assignment. The TAP is then formulated using QBTD travel time function and a modified path-based algorithm is developed to solve the UE problem. For clarity, a summary of the mathematical notations utilized in this section is provided below.

N	a set of nodes
A	a set of directed links(i,j), $i \in N$ and $j \in N$
O	a set of origin nodes
D	a set of destination nodes
r	an origin node, $r \in O$
s	a destination node, $s \in D$
d^{rs}	travel demand between origin r and destination s
Π^{rs}	a set of paths between origin r and destination s
π	a path, $\pi \in \Pi^{rs}$
h^{π}	flow on path π
Δt	time step
$x_{ij}(t)$	flow on link (i,j)
δ_{ij}^{π}	the number of times link (i,j) is used by path π, $\delta_{ij}^{\pi} = 1$ if path π uses link (i,j), and $\delta_{ij}^{\pi} = 0$ if it does not.
$tt_{ij}(Q_{ij}(t))$	time-dependent link travel time function
t_{ij}^f	free flow travel time on link (i,j)
$Q_{ij}(t)$	queue length on link (i,j) at time t
$w_{ij}(t)$	delay time on link (i,j) at time t
r_{ATMA}	a potential route traversing all links need to be maintained by ATMA vehicles
$C_{r_{\text{ATMA}}}$	the system cost of the route r_{ATMA}
$TSTT_0$	the TSTT without ATMA vehicles, following the UE principle
$TSTT_{r_{\text{ATMA}}}$	the TSTT following the UE principle and the selection of route r_{ATMA} by ATMA vehicles

3.1. Overview of the modeling framework

The selection of routes for ATMA vehicles results in different time-dependent capacity reduction patterns, which significantly affect the overall performance of the traffic system. In this study, we define the system cost of a particular route, denoted as $r_{\text{ATMA}} \in R$, for ATMA vehicles as the difference between the TSTT under the UE principle with and without ATMA operation, represented by Eq. (7).

$$C_{r_{\text{ATMA}}} = TSTT_{r_{\text{ATMA}}} - TSTT_0 \quad (7)$$

The optimal route, denoted as r_{ATMA}^*, can be obtained by minimizing the cost function:

$$r_{\text{ATMA}}^* = argmin\ (C_{r_{\text{ATMA}}}) \quad (8)$$

The overview of the modeling framework with bi-level objectives is depicted in Figure 5, where the higher-level objective is dedicated to determining the optimal route and

the lower-level is responsible for executing the traffic assignment process. In the higher-level, given a potential route r_{ATMA}, the time-dependent capacity reduction can be determined as analyzed in Section 2.2. The dynamic capacity pattern is then incorporated into the traffic assignment model in the lower level. As a result, the system cost, $C_{r_{\mathrm{ATMA}}}$, associated with that specific ATMA vehicle route is obtained for each iteration loop. If the cost $C_{r_{\mathrm{ATMA}}}$ is lower than the current minimum system cost C_{min}, both the minimum cost and the optimal route, r^{*}_{ATMA}, will be updated. The traffic assignment model incorporates the QBDT travel time function in the network loading module, using the given routes and route flows. The resulting time-dependent travel time and delays are utilized in the path set update and adjustment modules. The updated routes and route flows are then fed back into the network loading modules until the stop criteria is satisfied.

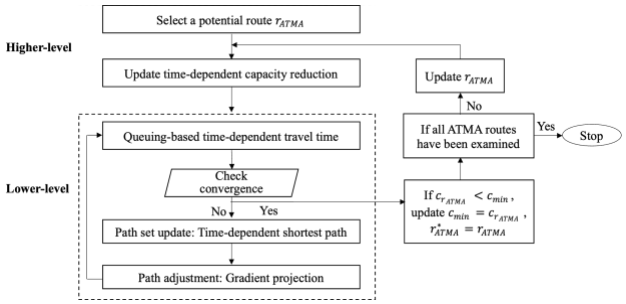


Figure 5. Overview of the modeling framework with bi-level objectives.

3.2. TAP Formulation and Solution Algorithm

A static UETAP is traditionally formulated using Beckmann's function [5], with link flows or path flows as the decision variables. The BPR function is commonly used as the link performance function; however, it cannot capture the dynamic queuing process. To address this limitation and achieve the lower-level objective, we propose a queuing-based TAP that incorporates dynamic characteristics by employing the QBTD travel time function, as analyzed in Section 2.3.

Before formulating this queuing-based TAP, two assumptions are made:

1) Each driver chooses the path between their origin and destination with the shortest travel time.
2) The time-dependent link flow $x_{ij}(t)$ is continuous and differentiate.

Consequently, we present the reformulated version of our queuing-based TAP as follows:

$$\min f(h,t) = \sum_{(i,j)\in A} \int_0^{\sum_{\pi\in\Pi^{\mathrm{rs}}} \delta_{ij}^{\pi} h^{\pi}} t_{ij}(x,t)\,dx \tag{9}$$

s. t.

$$x_{ij} = \sum_{\pi\in\Pi^{\mathrm{rs}}} \delta_{ij}^{\pi} h^{\pi}, \forall (i,j) \in A \tag{10}$$

$$\sum_{\pi\in\Pi^{rs}} h^{\pi} = d^{rs}, \forall r \in O, \forall s \in D \tag{11}$$

$$t_{ij}(x,t) = t_{ij}^{f} + w_{ij}\left(t + t_{ij}^{f}\right), \forall (i,j) \in A \tag{12}$$

$$w_{ij}\left(t + t_{ij}^{f}\right) = \frac{Q_{ij}(t + t_{ij}^{f})}{\mu}, \forall (i,j) \in A \tag{13}$$

$$Q_{ij}\left(t + t_{ij}^{f}\right) = \int_0^{t+t_{ij}^{f}} \left(x_{ij}(\tau) - \mu\right) d\tau, \forall (i,j) \in A \tag{14}$$

$$h^{\pi} \geq 0, \forall \pi \in \Pi^{\mathrm{rs}} \tag{15}$$

Eq. (9) is the objective function and Eqs. (10)~(15) serve as constraints. Eqs. (10) and (11) describe the link and path flow constraints. The QBTD travel time function of link ij at time t is defined in Eq. (12), and Eqs. (13) and (14) are time-dependent delay and queue function.

The UE traffic assignment was the solution to the optimization problem. By incorporating dynamic characteristics into the traditional static UETAP, the queuing-based TAP can achieve UE at each time interval using the gradient projection method, a path-based algorithms. Different from the conventional link performance function, such as the BPR function, the travel time function in Eq. (12) is dynamic and incorporates time t as a new variable. Consequently, the traditional gradient projection method is inapplicable for addressing this queuing-based TAP. As a result, adjustments are required in the gradient projection method to effectively solve the problem.

The time-dependent Beckmann function $f(h,t)$ is in terms of path flow h^{π} and its partial derivative with respect to any path flow h^{π} is given by Eq. (16).

$$\frac{\partial f(h,t)}{\partial h^{\pi}} = c^{\pi}(t) \tag{16}$$

Thus, the direction of the steepest descent is the negative gradient $-\nabla_{\mathbf{h}} f = -\mathrm{vect}\left(c^{\pi}(t)\right)$. We define the basic path to be a path $\hat{\pi}$ with the shortest travel time, and all the other paths are named nonbasic paths. Then the constraint of capacity in Eq. (11) can be transformed into Eq. (17) by eliminating the basic path flow variable.

$$h^{\hat{\pi}} = d^{rs} - \sum\nolimits_{\pi\in\Pi^{\mathrm{rs}}-\{\hat{\pi}\}} h^{\pi} \tag{17}$$

Substituting Eq. (17) into Eq. (9), the time-dependent Beckmann function becomes:

$$\hat{f}(h,t) = \sum_{(i,j)\in A} \int_0^{\delta_{ij}^{\hat{\pi}}\left(d - \sum_{\pi\in\Pi-\{\hat{\pi}\}} h^{\pi}\right) + \sum_{\pi\in\Pi-\{\hat{\pi}\}} \delta_{ij}^{\pi} h^{\pi}} t_{ij}(x,t)\,dx \tag{18}$$

The partial derivative of $\hat{f}(h,t)$ with respect to one of nonbasic path flow is now:

$$\frac{\partial \hat{f}(h,t)}{\partial h^{\pi}} = c^{\pi}(t) - c^{\hat{\pi}}(t) \tag{19}$$

Let Δh denote the amount of flow we shift away from a non-basis path π onto the basic path $\hat{\pi}$ at a time step size. The travel times on paths π and $\hat{\pi}$ after shifting are denoted by $c^{\pi}(\Delta h,t)$ and $c^{\hat{\pi}}(\Delta h,t)$. Our purpose is to choose a Δh so that the change in travel times, $g(\Delta h,t)$, is equal to zero at each time interval

$$g(\Delta h,t) = \sum_{(i,j)\in A} \left(\delta_{ij}^{\pi} - \delta_{ij}^{\hat{\pi}}\right) \cdot t_{ij}(x_{ij}(\Delta h),t) \tag{20}$$

So,

$$\frac{\partial g(\Delta h,t)}{\partial \Delta h} = -\sum_{(i,j)\in A_3 \cup A_4} \frac{\partial t_{ij}(\Delta h,t)}{\partial x_{ij}} \tag{21}$$

where $A_3 = \{(i,j) | \delta_{ij}^{\pi} = 1 \text{ and } \delta_{ij}^{\hat{\pi}} = 0\}$ and $A_4 = \{(i,j) | \delta_{ij}^{\pi} = 0 \text{ and } \delta_{ij}^{\hat{\pi}} = 1\}$.

Based on the assumption that the link flow is continuous and differentiated, Newton's method is adopted to estimate the shifted flow Δh at each time interval.

$$\Delta h = \frac{c^{\pi}(\Delta h,t) - c^{\hat{\pi}}(\Delta h,t)}{\sum_{(i,j)\in A_3 \cup A_4} \frac{\partial t_{ij}(\Delta h,t)}{\partial x_{ij}}} \tag{22}$$

The amount of flow to shift is then:

$$\Delta h = \min\{h_{\pi}, \frac{c^{\pi}(\Delta h,t) - c^{\hat{\pi}}(\Delta h,t)}{\sum_{(i,j)\in A_3 \cup A_4} \frac{\partial t_{ij}(\Delta h,t)}{\partial x_{ij}}}\} \tag{23}$$

Next, we need to derive $\frac{dt_{ij}}{dx_{ij}}$ to solve our proposed queuing-based TAP. If the problem is formulated in continuous time, we can derive the first derivation of travel time function to the link flow as Eq. (24).

$$\frac{\partial t_{ij}(x_{ij},t)}{\partial x_{ij}} = \frac{x_{ij}(t)}{\mu_{ij}(t)} \Big/ \frac{dx_{ij}(t)}{dt} \tag{24}$$

Since the link flow x_{ij} is unknown before we perform traffic assignment, it is impossible to derive the first derivation unless an exact function is given. Moreover, solving continuous-time models in large-scale networks could be a challenge, and hence numerical solution algorithms are usually developed based on discretized time intervals (e.g., Ziliaskopoulos [28], Qian, Shen [29], and Long, Chen [30]). To simplify the proximation process of the first derivation of $\frac{dt_{ij}}{dx_{ij}}$, we take discretized time interval in this manuscript. The analysis period is denoted as T, with a time interval set as Δt. Therefore, the total number of time intervals is $M = T/\Delta t$ with each interval $m = \{1,2,\dots,M\}$. We then assume that the flow on any link (i,j) remains constant (denoted as x_{ij}^{m}) during each mth time interval Δt. In this case, the queue length at m^{th} time interval becomes:

$$Q_{ij}(m\cdot\Delta t) = \int_{(m-1)\cdot\Delta t}^{m\cdot\Delta t} \left(x_{ij}^{m} - \mu_{ij}\right) d\tau \tag{25}$$

The time-dependent delay time can be calculated as:

$$w_{ij}(m\cdot\Delta t) = \frac{\int_{(m-1)\cdot\Delta t}^{m\cdot\Delta t} \left(x_{ij}^{m} - \mu_{ij}\right) d\tau}{\mu_{ij}} \tag{26}$$

Then the QBTD travel time function on link (i,j) is:

$$t_{ij}\left(x_{ij}, m\cdot\Delta t\right) = t_{ij}^{f} + \frac{\left(x_{ij}^{\,m+t_{ij}^{f}/\Delta t} - \mu_{ij}\right)\cdot\Delta t}{\mu_{ij}} \tag{27}$$

We can derive the $\frac{dt_{ij}}{dx_{ij}}$ as Eq. (28).

$$\frac{dt_{ij}\left(x_{ij}, m\cdot\Delta t\right)}{dx_{ij}} = \begin{cases} \dfrac{\Delta t}{\mu_{ij}(m\cdot\Delta t)}, & \text{if } x_{ij} > \mu_{ij} \\ 0, & \text{if } x_{ij} \le \mu_{ij} \end{cases} \tag{28}$$

Thus, we can obtain the amount of flow to shift in each time interval. To outline the procedure of the queuing-based traffic assignment, we first introduce some notations. For a total of N^{rs} origin-destination (OD) pairs, each OD pair is represented by $n \in \{1,2,\dots N^{rs}\}$ with a path set denoted as Π_n and individual paths within the set as $\pi_n \in \Pi_n$. The travel time for each path is denoted as c_{π_m}. The solution algorithm for this traffic assignment is detailed in the following.

Description of the solution algorithm

1. Initialization

Set the parameters:

- $m = 1, \widehat{\Pi}_n = \emptyset$
- $Q_{ij}(0) = 0, \mu_{ij}(0) = 0, d_n(0) = d_n$
- $x_{ij}(0) = 0, tt_{ij}(0) = t_{ij}^{f}, c_{\pi_n} = \sum_{(i,j)\in\pi_n} tt_{ij}$

2. Iteration Loop

2.1. Set $n = 1$, $t = m\cdot\Delta t$, update $\mu_{ij}(m\cdot\Delta t)$ and $d_n(m\cdot\Delta t)$

2.2. While Loop: $k = 1$

- Find the shortest path π_n^*, if not in $\widehat{\Pi}_n$ adding it.
- If there is only one path in $\widehat{\Pi}_n$, set the path flow $h_{\pi_n^*} = d_n(m \cdot \Delta t)$; otherwise, for each nonbasic path π_n calculate shift flow $\Delta h_{\pi_n} = \frac{c_{\pi_n} - c_{\pi_n^*}}{\sum \frac{dtt_{ij}}{d_{x_{ij}}}}$. Set $h_{\pi_n^*}^k = h_{\pi_n^*}^{k-1} + \sum \Delta h_{\pi_n}$, and $h_{\pi_n}^k = h_{\pi_n}^{k-1} - \Delta h_{\pi_n}$. Remove path π_n if $h_{\pi_n}^k \leq \varepsilon$

2.3. Update link flow $x_{ij}^k(m \cdot \Delta t)$

2.4. Calculate link travel time $tt_{ij}^k(m \cdot \Delta t) = t_{ij}^f + \frac{\max(0, Q_{ij}((m-1)\cdot\Delta t) + \Delta t \cdot (x_{ij}^k(m\cdot\Delta t) - \mu_{ij}((m-1)\cdot\Delta t))}{\mu_{ij}((m-1)\cdot\Delta t)}$

2.5. Update path cost $c_{\pi_n}^k = \sum_{(i,j)\in\pi_n} tt_{ij}^k(m \cdot \Delta t)$

3. Convergence check

3.1. If $\gamma \leq \epsilon$ or $k \geq iter_{max}$:
- If $n > N^{rs}$:
  - If $t > T$: output $TSTT$ and stop
  - Else: update $Q_{ij}(m \cdot \Delta t), m = m + 1$, return to 2.1
- Else: update n= $n + 1$, return to 2.1

3.2. Else: $k = k + 1$ return to 2.2

4. Numerical Experiments

To validate and assess the queuing-based traffic assignment approach, we conducted numerical experiments on two different network sizes: a small-size simple network and the Sioux Falls network [31]. Initially, we verified the convergence pattern of our proposed algorithm and assessed the benefits of incorporating capacity drop modeling and the QBTD travel time function in UE traffic assignment. Additionally, we utilized the queuing-based approach to analyze the impact of ATMA vehicles on the overall traffic system and identify the optimal route with the lowest system cost. Sensitivity analysis was also performed to examine how the results vary with changes in demand and the travel speed of ATMA vehicles. The experiments were conducted using Path4GMNS [32], an open-source platform for UE traffic assignment, which was adapted and modified to accommodate our queuing-based TAP.

4.1. Methodology Validation and Benefit Analysis

Figure 6 illustrates the topology of the small-size network. The traffic demand from the origin node 1 to the destination node 4 is 6,000 veh/hr. Two paths are available: P1 (1,4) and P2 (1,2,3,4). For each link, the free flow travel speed is 40 mph, and the backward wave speed is 12 mph. Additionally, the capacity of each link is 3,000 veh/hr. The FFTT of first path is 90 s, and it is 135 s for the second path.

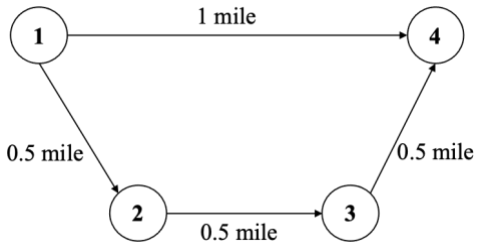


Figure 6. The simple network topology.

The convergence pattern of the adjusted gradient projection algorithm was verified using various time intervals, ranging from 5 s to 1 min, and the results are shown in Figure 7. Notably, we observed that it requires 7 iterations to converge to 0.01% when the time interval is set to 5 s, while it requires 28 iterations if the time interval is 1 min. As the time interval increases, the initial relative gap also increases, consequently necessitating more iterations to achieve convergence.

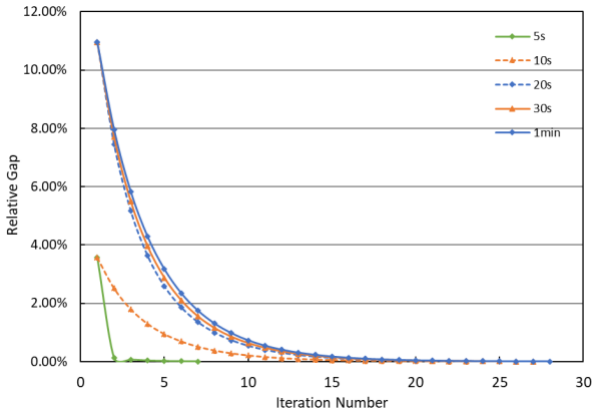


Figure 7. Convergence pattern.

Next, we conducted experiments to evaluate the performance of the proposed queuing-based traffic assignment approach. For each time interval, we employed the adjusted gradient projection method to perform UE traffic assignment. In this algorithm, we incorporated the QBTD travel time function, considering capacity drop. To assess the benefits of our proposed model, we established two common-used benchmark models.

- *Capacity drop-ignored model*: capacity drop was ignored in the QBTD travel time function during UE traffic assignment. However, realistic capacity was considered to determine actual travel time and relative gap. By comparing the proposed model with capacity drop-ignored model, we could analyze the benefits of capacity drop modeling.
- *BPR-based model*: the BPR travel time function considering capacity drop was employed for UE traffic assignment in each time interval. While the QBTD

travel time function, accounting for capacity drop, was used to calculate the actual travel time and relative gap. By comparing the proposed model with the BPR-based model, we could observe the advantages of using the QBTD travel time function.

In the experiment, the total time was set to 600 s with a time interval of 30 s. We assumed that the link (1,4) required maintenance, which was carried out by slow-moving ATMA vehicles with a speed of 3.5 m/s. The total maintenance time was 460 s, during which the effective discharge rate on link (1,4) decreased. Once the maintenance was completed, the link's capacity returned to normal. For each time interval, a UE traffic assignment was performed, and the queuing length and travel time of the link were updated accordingly.

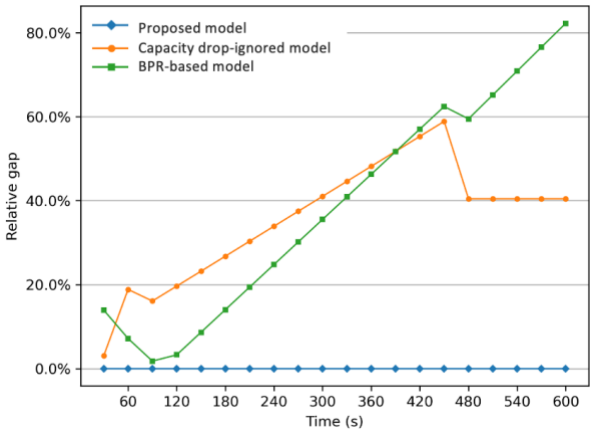


Figure 8. Relative gap comparison of the proposed model and two benchmark models.

Figure 8 presents a comparison of the relative gaps between the proposed model and the two benchmark models. The blue line represents the relative gap of the proposed model, which remains close to 0.0%. It is observed that the corrected relative gap of capacity drop-ignored model (depicted by the orange line) gradually increases to 60% during the capacity reduction period and then drops to around 40% after capacity returns to normal. This implies that disregarding the capacity drop in the UE traffic assignment results in a path flow mismatch, deviating from UE in this simple network. Furthermore, when the static BPR travel time function is used in the UE traffic assignment, the corrected relative gap (illustrated by the green line) exhibits a continuing increase, reaching up to 82%. It indicates that if the congested condition is not accurately captured, the resulting path flow significantly deviates from the equilibrium.

4.2. Routing Plan Benefit Analysis

This section focuses on examining the benefits of the proposed model in the routing plan for ATMA vehicles. In a network, multiple links may require simultaneous maintenance, which poses a challenge in finding the optimal route for ATMA vehicles. Typically, the shortest path is considered the best choice. However, the slow-moving ATMA vehicles become a moving bottleneck, leading to capacity drops. Consequently, the link queue length and travel time increase during the maintenance period. For different routes, it is necessary to determine the time-varying capacity for specific links. Our proposed model incorporates capacity reduction in UE traffic assignment and quantifies the impact of various ATMA routes on the traffic system. The obtained results will be compared with those of the two benchmark models to evaluate the effectiveness of our approach.

4.2.1. Convergence Pattern

The Sioux Falls network, consisting of 24 nodes and 75 links, is used to illustrate the impact of different routes, as depicted in Figure 9. The free flow travel speed is 60 mph, while the backward wave speed is 20 mph. In our analysis, we will focus on the maintenance of four specific links: (6,8), (16,17), (15,22), and (11,14), which are represented by red links in Figure 9. The travel speed of ATMA vehicles during maintenance is set to 10 mph. There are multiple routes available for ATMA vehicles with different origins and destinations. In this study, we select node 6 as the origin and node 14 as the destination. Specifically, we consider the impact of the shortest 10 paths on TSTT. For instance, the FFTT for general traffic of the shortest path is 38 min, whereas for ATMA vehicles, it increases to 228 min. The FFTT for the other nine paths ranges from 39 min to 48 min for general traffic, and from 23 4min to 288 min for ATMA vehicles.

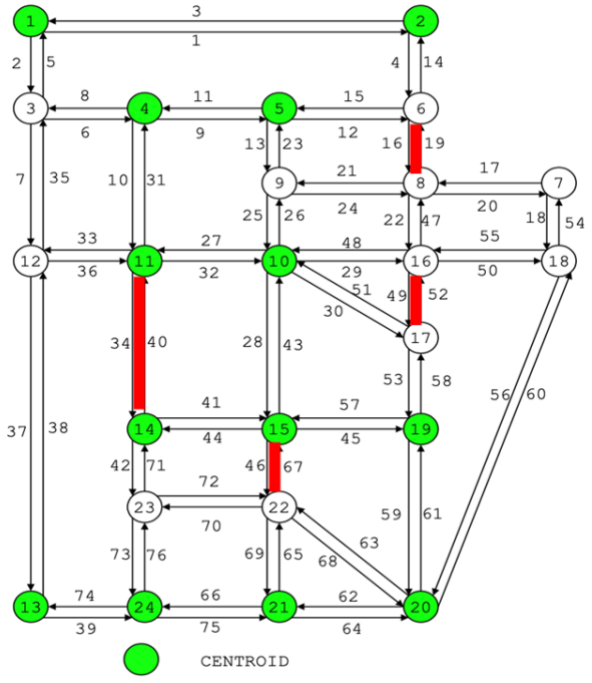


Figure 9. Sioux Falls Test Network.

The experiment was conducted over a total duration of 5 hours, with a time interval of 5 s, resulting in 3,600 intervals in total. Convergence criteria were established, requiring either a maximum of 20 iterations or a relative gap below 0.1% for each time interval. During our experimental analysis, we thoroughly investigated the ten different paths characterized by varying capacity drop patterns. The computational time was 46 s and the results exhibited an impressive level of convergence, with relative gaps below 0.1% observed in 98.9% to 99.1% of all time intervals across the examined paths. For instance, Figure 10 presents a cumulative histogram of the relative gap. These results demonstrate the effectiveness of the proposed queueing-based traffic assignment and the modified path-based algorithm in achieving convergence.

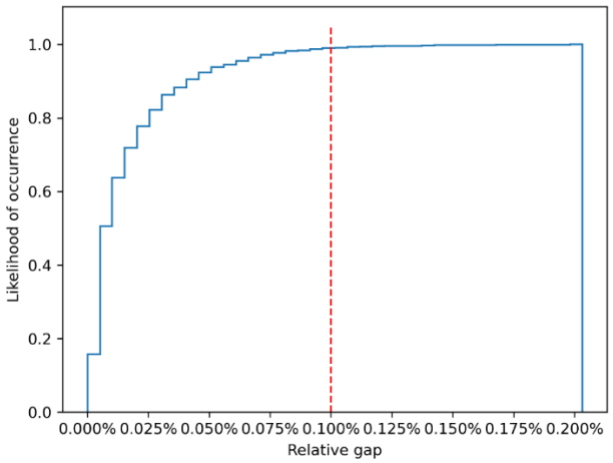


Figure 10. Cumulative histogram of the relative gap.

4.2.2. Benefit Analysis

To evaluate the benefits of the capacity drop modeling and the QBTD travel time function, we conducted experiments for each path using the proposed model and two benchmark models. Figure 11 presents cumulative histograms comparing the corrected relative gaps of the benchmark models with those of the proposed model. The results reveal significant deviations from the UE in capacity drop-ignored model, with corrected relative gaps exceeding 0.1% observed in 90.4% to 99.1% of all time intervals across the examined paths when the capacity reduction is disregarded. Additionally, when the UE traffic assignment utilizes the BPR function instead of the QBTD travel time function, the results demonstrate substantial deviations from the UE, with corrected relative gaps exceeding 2% observed in 98.9% of all time intervals across the examined paths. This highlights the inadequacy of the static BPR travel time function in capturing time-varying information and its consequent significant deviation from the UE.

Table 1 provides a summary of the average corrected relative gaps for the 10 paths between the proposed model and the two benchmark models. The average relative gap for the proposed model is approximately 0.019%. However, if the capacity reduction is disregarded in the UE traffic assignment process, the average relative gap increases to a range of 0.286% to 0.582%. Additionally, when using the static BPR travel time function, the average relative gap sharply increases to a range of 11.71% to 11.86%. These results highlight the substantial impact of considering capacity drop and utilizing the QBTD travel time function in achieving a closer approximation to the UE.

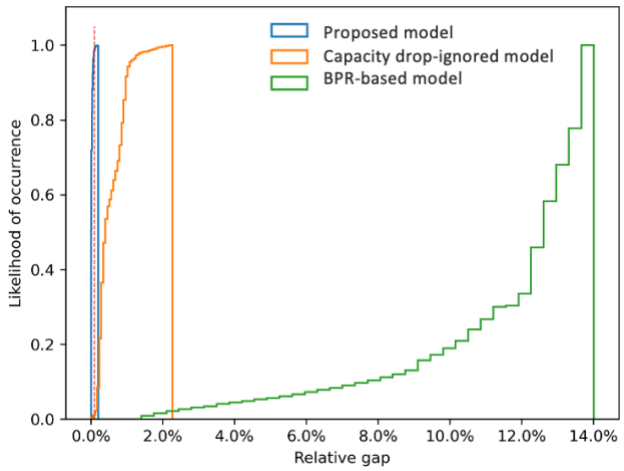


Figure 11. Cumulative histogram of the corrected relative gap compared with benchmark models.

Table 1. Comparison of Average Relative Gap

	Path 1	Path 2	Path 3	Path 4	Path 5	Path 6	Path 7	Path 8	Path 9	Path 10
Proposed model	0.019%	0.019%	0.019%	0.018%	0.019%	0.019%	0.019%	0.018%	0.019%	0.019%
Capacity drop-ignored model	0.573%	0.448%	0.307%	0.405%	0.582%	0.323%	0.286%	0.410%	0.408%	0.526%
BPR-based model	11.71%	11.71%	11.84%	11.78%	11.72%	11.86%	11.80%	11.79%	11.73%	11.75%

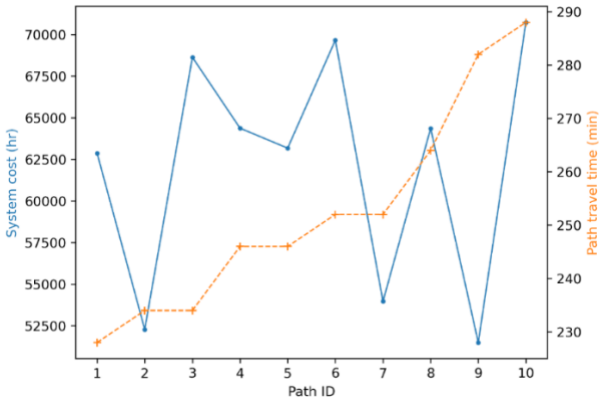


Figure 12. System cost and travel time of ten paths.

After analyzing the benefits of the proposed model, we proceed to investigate the impact of different ATMA routes on the traffic system cost. The initial TSTT without ATMA vehicles is recorded as 5,656,962 hours. However, with the introduction of ATMA vehicles for maintenance, the TSTT experiences a noticeable increase. Figure 12 illustrates the increased TSTT compared to the TSTT in the absence of ATMA maintenance and the travel time of different paths. It is observed that as the path travel time increases, the additional TSTT does not necessarily increase proportionally. For instance, Path 10 results in TSTT having the largest increase of 1.25%. On the other hand, paths 2, 7, and 9 have relatively lower impacts on the total traffic system, for example, Path 9 yields the smallest increase in TSTT of 0.91%. However, the shortest path, Path 1, exerts a higher impact on the total traffic system, resulting in an increase of 1.11% in TSTT.

4.3. Sensitivity Analysis

In this section, a sensitivity analysis of the Sioux Falls network is conducted to explore the changes in the total traffic system's impact when traffic demand and the speed of ATMA vehicles are altered.

4.3.1. When Demand Changes

The demand for each OD pair is set to increase by 10%, 20%, and 30%, respectively, to investigate the impact on total traffic system. The TSTT experiences a significant increase without ATMA vehicles as the demand rises. For instance, when the demand increases by 10%, the TSTT increases by 38%. Furthermore, with a 30% increase in demand, the TSTT shows a substantial rise of 130%. This demonstrates the high sensitivity of the TSTT to changes in demand.

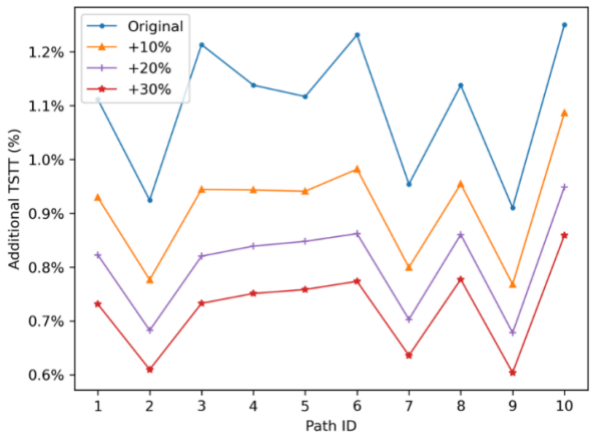


Figure 13. Additional TSTT compared with no ATMA when demand changes.

Figure 13 presents the additional TSTT (or the system cost) with changing demand, illustrating that the rate of TSTT increase diminishes as demand rises. When comparing the TSTT with no ATMA vehicles, the increase in TSTT ranges from 0.77% to 1.09% for a 10% increase in demand. In the case of a 20% increase in demand, the increase in TSTT ranges from 0.68% to 0.95%. Similarly, for a 30% increase in demand, it ranges from 0.60% to 0.86%. The system costs still exhibit significant variation among different routes of ATMA vehicles. Additionally, it is worth noting that paths 9 and 2 have minimal impact on system traffic, indicating that the optimal route for ATMA vehicles remains relatively unaffected by changes in demand.

4.3.2. When ATMA speed change changes

We also conducted experiments to examine the impact of varying the speed of ATMA vehicles on the traffic system. Specifically, we considered speeds of 10 mph, 15 mph, and 20 mph, and analyzed the additional TSTT for ten paths in comparison to scenarios without ATMA vehicles or capacity reduction. Figure 14 illustrates the additional TSTT with changing travel speeds. When the speed of ATMA vehicles is set at 10mph, we observe an increase in TSTT ranging from 0.91% to 1.25%. However, as the speed increases to 15 mph and 20 mph, the increase in TSTT ranges from 0.49% to 0.68% and 0.29% to 0.41%, respectively. These findings suggest that the rate of TSTT increase diminishes as the speed of ATMA vehicles increases. As the speed of ATMA vehicles increases, the disparity in system costs diminishes across various routes. We notice that paths 9 and 2 still have minimal impact on the system traffic, which indicates that the optimal route for ATMA vehicles remains relatively unaffected by variations in travel speed.

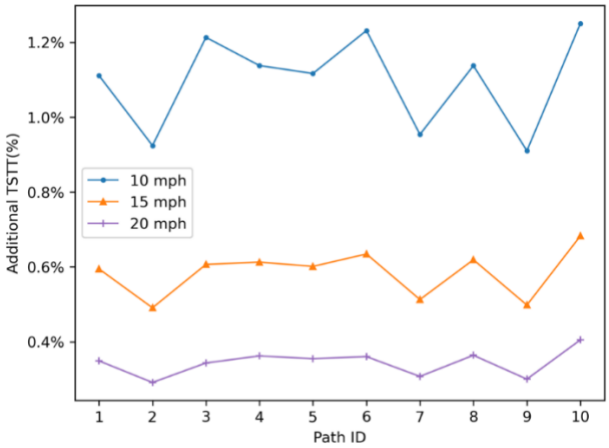


Figure 14. Additional TSTT compared with no ATMA when demand changes.

5. Conclusion

This study proposed a queuing-based traffic assignment approach to optimize ATMA vehicle deployment in transportation networks. By introducing a QBTD travel time function to account for capacity drop, dynamic characteristics were incorporated into the static UETAP. A reformulated TAP by integrating the QBTD travel time function, and a modified path-based algorithm was developed to solve the model and achieve UE. Validation experiments conducted on both small-scale and larger-scale networks demonstrated consistent convergence pattern. The results were compared with two benchmark models to highlight the benefits of time-varying capacity drop modeling and the QBTD travel time function. The methodology was further applied to a case study for ATMA deployment, with sensitivity analysis evaluating the effects of varying traffic demand and ATMA operating speeds. The key findings include:

- Ignoring capacity drop led to significant deviations from UE, with an average corrected relative gap increasing from 0.02% to 0.43% in the large-scale network. Using the static BPR travel time function caused even greater deviations, with an average relative gap of 11.78%.
- The shortest path was not always the optimal route for ATMS vehicles, as it does not account for the effects of time-varying capacity reduction.
- System costs varied significantly across ATMA routes under increased traffic demand. However, as the operating speed of ATMA vehicles increased, disparities in system costs diminished, suggesting a strategy to reduce traffic impacts.

This research provides a practical framework for modeling and optimizing ATMA deployment, offering insights for improving traffic flow and minimizing disruptions. Despite its contributions, this study has limitations that warrant future investigation:

- The model assumes constant traffic flow within each time interval, which may oversimplify real-world traffic dynamics. Future work could incorporate more realistic and variable flow patterns.
- Although the methodology was validated on both small-scale and large-scale networks, applying it to very large and complex networks may require additional computational enhancements.

Declarations

Conflict of interest On behalf of all the authors, the corresponding author states that there is no conflict of interest.

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